LECTURES NOTES

IN

(WEB PROGRAMMING I)

FOR SECOND LEVEL

IT DEPARTMENT

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CHAPTER 1

INTRODUCTION

1.1. The Internet

Developed beginning in the 1960s, sponsored by ARPA (Dept. of Defense Advanced Research Projects Agency). Protocol for communication between computers is called TCP/IP: Transmission Control Protocol / Internet Protocol.

Information is transmitted by *packet switching:* data is broken up into packets, which are then passed from one computer to the next until they reach their destination.

IP requires each connected computer to have a unique *address:* a number such as 150.108.64.52. Clients such as home computers are generally assigned IP addresses dynamically (by DHCP) from a pool of available numbers upon connecting to the Internet. Server computers have static IP addresses.

Computers can also be identified by names. Domain Name Service (DNS) translates between name and number. Internet names have two parts: a host name (the first component) and a domain (everything else). Example: erdos.dsm.fordham.edu. Various protocols are built on top of TCP/IP:

- E-mail (SMTP)
- File Transfer Protocol (FTP)
- Drive and printer sharing protocols, e.g. NetBIOS
- Network Time Protocol (NTP) for setting computers' clocks.
- Hypertext Transport Protocol (HTTP): the World-Wide Web

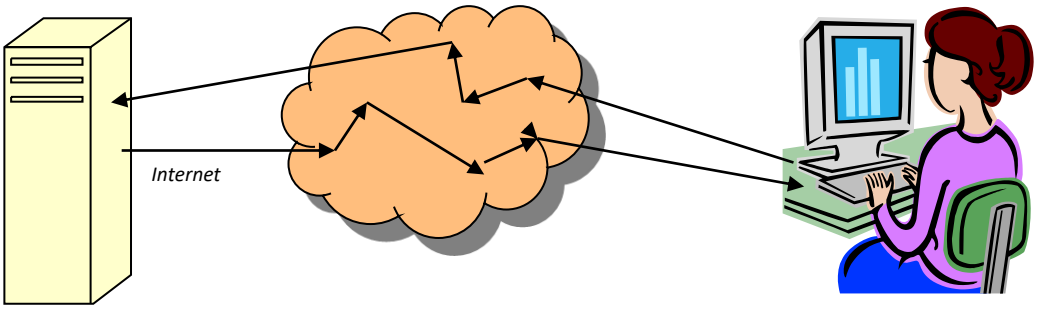
TCP requires computers to connect via numbered *ports* defined by the network software applications. Each protocol uses a different port. For instance, Web transactions usually use port 80. Note that the Web is just one of many protocols using the Internet.

1.2. The World-Wide Web

Around 1990 by Tim Berners-Lee, a programmer at CERN (a physics lab in Geneva) to simplify sharing of information between research groups.

Based on client-server model:

Client computer runs a browser (e.g. Internet Explorer or Netscape) that requests information from a server. Server computer runs a web server (e.g. Microsoft Internet Information Server or Apache Web Server) that listens for requests and sends back information



1.3. Web Programming

There are two broad divisions of web development

- front-end development (also called client-side development) and
- back-end development (also called server-side development)

1.3.1. Front-end development

Refers to constructing what a user sees when they load a web application (the content, design and how you interact with it), this is done with three codes

- HTML, short for **Hyper Text Markup Language**,
  - is a special code for 'marking up' text in order to turn it into a web page. Every web page on the net is written in HTML, and it will form the backbone of any web application.
- CSS, short for **Cascading Style Sheets**,
  - is a code for setting style rules for the appearance of web pages. CSS handles the cosmetic side of the web.
- Finally, **JavaScript**
  - is a scripting language that's widely used to add functionality and interactivity to web pages.

1.3.2. Back-end development

Controls what goes on behind the scenes of a web application. A back-end often uses a database to generate the front-end. Back-end scripts are written in many different coding languages and frameworks, such as…

- **PHP**
- **Ruby on Rails**
- ASP.NET
- Perl

- **Java**
- Node.js
- **Python**

1.4. Web Design

Is the visual and page layout of a website. It goes hand-in-hand with web development in the creation of a static website or dynamic web application. The design is the first thing people notice when they arrive on a website, and if it's not good enough, they'll leave.

The web design process starts with a visual concept, which you could sketch by hand or with software like Photoshop. Then, you use **HTML** and **CSS** to build the website.

- HTML handles the basic structure and 'bones' of your page,
- while CSS handles the style and appearance.

If you're a good web designer, you'll also pay attention to concepts like:

- **Responsive design** is a popular technique for making websites look good and function well on additional devices, like phones and tablets. It does this by setting different CSS rules for different browser widths
- **Aesthetics** plays a big role in web design, since it's all about how web pages look.
- **Usability** is another important component of web design. This is all about how intuitive a website is, and how easy it is for users to find or do what they want.

- **Accessibility** – a very important web design practice. It is all about making web pages viewable by people that use assistive devices to browse the web.

CHAPTER 2

HTML

2.1. What is HTML?

HTML stands for Hypertext Markup Language, and it is the most widely used language to write Web Pages.

Hypertext refers to the way in which Web pages (HTML documents) are linked together. Thus, the link available on a webpage is called Hypertext.

As its name suggests, HTML is a Markup Language which means you use HTML to simply "mark-up" a text document with tags that tell a Web browser how to structure it to display.

Originally, HTML was developed with the intent of defining the structure of documents like headings, paragraphs, lists, and so forth to facilitate the sharing of scientific information between researchers. Now, HTML is being widely used to format web pages with the help of different tags available in HTML language.

2.2. HTML Editor

A word processor that has been specialized to make the writing of HTML documents more effortless. There are many different programs that you can use to create web documents.

HTML Editors enable users to create documents quickly and easily by pushing a few buttons. Instead of entering all of the HTML codes by hand. These programs will generate the HTML Source Code for you.

HTML Editors are excellent tools for experienced web developers; however; it is important that you learn and understand the HTML language so that you can edit code and fix “bugs” in your pages. We will focus on using the standard Microsoft Windows text editors, NotePad.

2.3. HTML Tags

As told earlier, HTML is a markup language and makes use of various tags to format the content. These tags are enclosed within angle braces <Tag Name>. Except few tags, most of the tags have their corresponding closing tags. For example, <html> has its closing tag</html> and <body> tag has its closing tag </body> tag etc.

To learn HTML, you will need to study various tags and understand how they behave, while formatting a textual document. Learning HTML is simple as users have to learn the usage of different tags in order to format the text or images to make a beautiful webpage.

2.4. Creating a Basic Starting Document

A typical HTML document will have the following structure:

```
<html>
  <head>
    Document header related tags
  </head>
  <body>
    Document body related tags
  </body>
</html>
```

It is like the head of a human, because it is used to identify the file. head tag should lie inside "html" tag. head tag has many purposes.

But at this point we will discuss only one of its tags <title>. Title this tag is used to set the title of the page. The title will be displayed on the left top corner of the browser page.

```
<html>
<head>
<title> My Page
</title>
</head>
</html>
```

BODY

It is like the body of a human. BODY tag should lie inside "html" tag. Anything we type inside the body tag will be displayed in the browser page. Document properties are controlled by attributes of the BODY element. For example, there are color settings for the background color of the page, the document's text and different states of links.

```
<html>
<head>
<title> My Page
</title>
</head>
<body>
This is my first page
 </body>
</html>
```

2.5. HTML Color Coding Methods

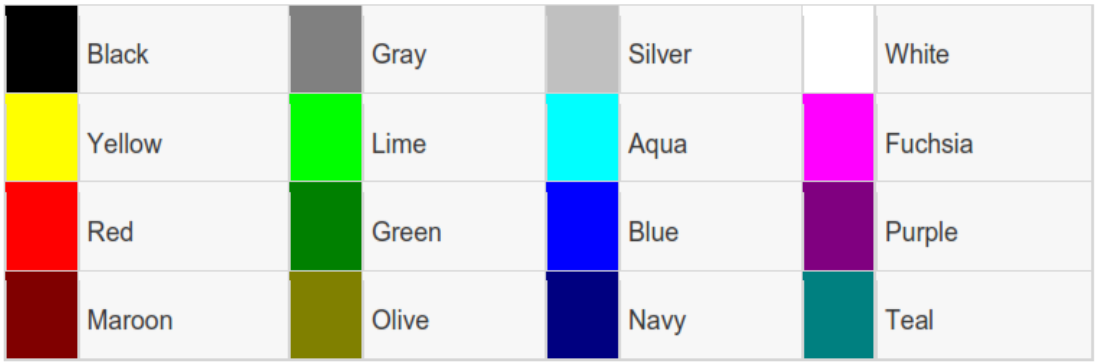
There are following three different methods to set colors in your web page:

- **Color names -** You can specify color names directly like green, blue or red.
- **Hex codes -** A six-digit code representing the amount of red, green, and blue that makes up the color.
- **Color decimal or percentage values -** This value is specified using the rgb() property.

HTML Colors - Color Names

You can specify direct a color name to set text or background color. W3C has listed 16 basic color names that will validate with an HTML validator but there are over 200 different color names supported by major browsers.

Here is the list of W3C Standard 16 Colors names and it is recommended to use them.



Example:

```
<html>
```

```
<head>
<title>HTML Colors by Name</title>
</head>
<body text="blue" bgcolor="green">
<p>Use different color names for body and table and see the result.</p>
</body>
</html>
```

HTML Colors - Hex Codes

Colors are represented as hexadecimal values. The color value start with symbol # and six digits (RRGGBB) first two digits from right represent blue color has values from 00 to FF. Second two digits represent green color has values from 00 to FF. Third two digits represent red color has values from 00 to FF. For example, Yellow color represent by #FFFF00, that means the yellow color combination between red color and green color and no blue color> the following table show some colors and its codes:

Color	Hex-Code
1. WHITE	1. #FFFFFF
2. BLACK	2. #000000
3. RED	3. #FF0000
4. GREEN	4. #00FF00
5. BLUE	5. #0000FF
6. MAGENTA	6. #FF00FF
7. CYAN	7. #00FFFF
8. YELLOW	8. #FFFF00
9. AQUAMARINE	9. #70DB93

10.BAKER’S CHOCOLATE	10.#5C3317
11.VIOLET	11.#9F5F9F
12.BRASS	12.#B5A642
13.COPPER	13.#B87333
14.PINK	14.#FF6EC7
15.ORANGE	15.#FF7F00

HTML Colors - RGB Values

This color value is specified using the rgb( ) property. This property takes three values, one each for red, green, and blue. The value can be an integer between 0 and 255 or a percentage.

Note: All the browsers does not support rgb() property of color so it is recommended not to use it.

```
<html>
<head>
<title>HTML Colors by Name</title>
</head>
<body text="rgb(0,0,255)" bgcolor="rgb(0,255,0)">
<p>Use different color names for body and table and see the result.</p>
</body>
</html>
```

2.6. The Body Element

The BODY element of a web page is an important element in regards to the page's appearance. Here are the attributes of the **BODY** tag to control all the levels:

TEXT="#RRGGBB" to change the color of **all the text** on the page (**full page text color.**)

Background Color

This element contains information about the page's background color, the background image, as well as the text and link colors.

It is very common to see web pages with their background color set to white or some other colors.

To set your document's background color, you need to edit the <BODY> element by adding the BGCOLOR attribute. The following example will display a document with a white background color:

<BODY BGCOLOR="#FFFFFF"></BODY>

TEXT Color

The TEXT attribute is used to control the color of all the normal text in the document. The default color for text is black. The TEXT attribute would be added as follows:

<BODY BGCOLOR="#FFFFFF" TEXT="#FF0000"></BODY>

In this example the document's page color is white and the text would be red

LINK, VLINK, and ALINK

These attributes control the colors of the different link (that moves from page to page or any elements in the page) states:

1. **LINK** – initial appearance of the link that means the link has not been visited – default = Blue.
2. **VLINK** – the link color change when that has been visited – default = Purple.
3. **ALINK** – active link being clicked – default = Yellow.

The Format for setting these attributes is:

<BODY BGCOLOR="#FFFFFF" TEXT="#FF0000"
LINK="#0000FF"

VLINK="#FF00FF"

ALINK="FFFF00"> </BODY>

Using Image Background

The BODY element also gives your ability of setting an image as the document's background. You can use BACKGROUND attribute by send image file name to it (note: the image file location must be in the same folder of your html file if you write the image file name without file path). An example of a background image's HTML code is as follows:

<BODY BACKGROUND="hi.gif" BGCOLOR="#FFFFFF">
</BODY>

Test Example:

```
<html>
<head>
```

```
<title> My First Page
</title>
</head>
<BODY BACKGROUND="test.gif" BGCOLOR="#FFFF00"
TEXT="#FF0000" LINK="#0000FF" VLINK="#FF00FF"
ALINK="FFFF00">
This is my first page
</body>
</html>
```

Once you have created your basic starting document and set your document properties it is a good idea to save your file. To save a file, in NotePad, follow these steps:

1. Locate and click on the menu called "File".
2. Select the option under File Menu labeled "Save As".
3. In the "File Name" text box, type in the entire name of your file (including the extension name .html).

CHAPTER 3

HTML Text Prosperities

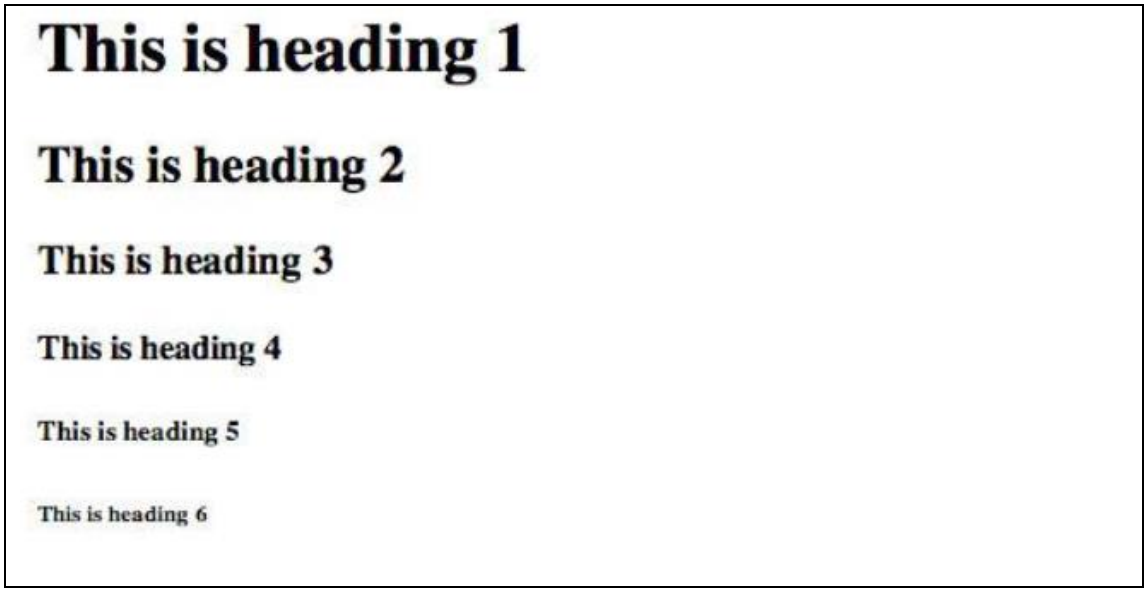
3.1. Heading Tags

Any document starts with a heading. You can use different sizes for your headings. HTML also has six levels of headings, which use the elements <h1>, <h2>, <h3>, <h4>, <h5>, and <h6>. H1: should be used as the highest level of heading, H2 as the next highest, and so forth. While displaying any heading, browser adds one line before and one line after that heading.

Example:

```
<html>
<head>
<title>Heading Example</title>
</head>
<body>
<h1>This is heading 1</h1>
<h2>This is heading 2</h2>
<h3>This is heading 3</h3>
<h4>This is heading 4</h4>
<h5>This is heading 5</h5>
<h6>This is heading 6</h6>
</body>
</html>
```

This will produce the following result:



3.2. Paragraph Tag

The tag offers a way to structure your text into different paragraphs. Each paragraph of text should go in between an opening and a closing tag

**Line Break:
**

This is a tag which is used to break the line and move to next line.

This tag does not require closing tag. Whenever you want the text to start in a new line give the tag
.

Example:

```
<HTML>
<HEAD>
<TITLE> Example Page</TITLE>
</HEAD>
<BODY>
<H1> Heading 1 </H1>
<P>Paragraph 1, <BR>
```

```
Line 2 <BR> Line 3 <BR>….
</P>
</BODY>
</HTML>
```

The output:

Heading 1

Paragraph 1,….

Line 2

Line 3

….

Horizontal Rule, <HR>

The <HR> element causes the browser to display a horizontal line (rule) in your document. <HR> does not use a closing tag, </HR>.

Attribute	Description	Default Value
SIZE	Height of the rule in pixels	2 pixels
WIDTH	Width of the rule in pixels or percentage of screen width	100%
NOSHADE	Draw the rule with a flat look instead of a 3D look	Not set (3D look)
ALIGN	Aligns the line (Left, Center, Right)	Center
COLOR	Sets a color for the rule (IE 3.0 or later)	Not set

**bold font: **

Any text given in between the "b" tag will appear as bold

underline tag: <u></u>

Any text given in between the "u" tag will appear underlined

**italics: **

Any text given in between the "i" tag will appear in italics

** :**

Any text given in between the "center" tag will be aligned center

Example:

```
<html>
<head>
<title> My Page
</title>
</head>
<body bgcolor="green" background="./test.jpg">
<center> <b> <u> <i>This is my first page </i> </u> </b> </center>
</body>
</html>
```

Strike Text

Anything that appears within <strike>...</strike> element is displayed with strikethrough.

Monospaced Font

The content of a <tt>...</tt> element is written in monospaced font. Most of the fonts are known as variable-width fonts because different letters are of different widths (for example, the letter 'm' is wider than the letter 'i'). In a monospaced font, however, each letter has the same width.

Superscript Text

The content of a ^{...} element is written in superscript; the font size used is the same size as the characters surrounding it but is displayed half a character's height above the other characters.

Subscript Text

The content of a _{...} element is written in subscript; the font size used is the same as the characters surrounding it, but is displayed half a character's height beneath the other characters.

Example:

```
<html>
<head>
<title>Monospaced Font Example</title>
</head>
<body>
<p>The following word uses a <strike>strikethrough</strike> typeface.</p>
<p>The following word uses a <sup>superscript</sup> typeface.</p>
<p>The following word uses a <sub>subscript</sub> typeface.</p>
<p>The following word uses a <tt>monospaced</tt> typeface.</p>
```

```
</body>
</html>
```

The output:

The following word uses a ~~strikethrough~~ typeface.

The following word uses a `monospaced` typeface.

The following word uses a ^{superscript} typeface.

The following word uses a $_{subscript}$ typeface

Inserted Text

Anything that appears within <ins>...</ins> element is displayed as inserted text.

Deleted Text

Anything that appears within ~~...~~ element, is displayed as deleted text.

Larger Text

The content of the <big>...</big> element is displayed one font size larger than the rest of the text surrounding

Smaller Text

The content of the <small>...</small> element is displayed one font size smaller than the rest of the text surrounding.

```
<html>
<head>
<title>Deleted Text Example</title>
```

```
</head>
<body>
<p>I want to drink <del>cola</del> <ins>wine</ins></p>
<p>The following word uses a <big>big</big> typeface.</p>
<p>The following word uses a <small>small</small> typeface.</p>
</body>
</html>
```

The output:

I want to drink ~~cola~~ wine

The following word uses a big typeface.

The following word uses a small typeface.

3.3. HTML Font

Fonts play a very important role in making a website more user friendly and increasing content readability. Font face and color depends entirely on the computer and browser that is being used to view your page but you can use HTML tag to add style, size, and color to the text on your website. You can use a <basefont> tag to set all of your text to the same size, face, and color.

The font tag is having three attributes called size, color, and face to customize your fonts. To change any of the font attributes at any time within your webpage, simply use the tag. The text that follows will remain changed until you close with the tag. You can change one or all of the font attributes within one <font> tag.

Set Font Size

You can set content font size using size attribute. The range of accepted values is from 1(smallest) to 7(largest). The default size of a font is 3.

```
<html>
<head>
<title>Setting Font Size</title>
</head>
<body>
<font size="1">Font size="1"</font><br />
<font size="2">Font size="2"</font><br />
<font size="3">Font size="3"</font><br />
<font size="4">Font size="4"</font><br />
<font size="5">Font size="5"</font><br />
<font size="6">Font size="6"</font><br />
<font size="7">Font size="7"</font>
</body>
</html>
```

The output:

Font size="1"

Font size="2"

Font size="3"

Font size="4"

Font size="5"

Font size="6"

Font size="7"

Relative Font Size

You can specify how many sizes larger or how many sizes smaller than the preset font size should be. You can specify it like <font size="+n"> or <font size="-n">

Example

```
<html>
<head>
<title>Relative Font Size</title>
</head>
<body>
<font size="-1">Font size="-1"</font><br />
<font size="+1">Font size="+1"</font><br />
<font size="+2">Font size="+2"</font><br />
<font size="+3">Font size="+3"</font><br />
<font size="+4">Font size="+4"</font>
</body>
</html>
```

This will produce the following result:

Font size="-1"

Font size="+1"

Font size="+2"

Font size="+3"

Font size="+4"

Setting Font Face

You can set font face using face attribute but be aware that if the user viewing the page doesn't have the font installed, they will not be able to see it. Instead user will see the default font face applicable to the user's computer.

Example

```
<html>
<head>
<title>Font Face</title>
</head>
<body>
<font face="Times New Roman" size="5">Times New Roman</font><br />
<font face="Verdana" size="5">Verdana</font><br />
<font face="Comic sans MS" size="5">Comic Sans MS</font><br />
<font face="WildWest" size="5">Aharoni</font><br />
<font face="Bedrock" size="5">Castellar</font><br />
</body>
</html>
```

This will produce the following result:

Times New Roman

Verdana

Comic Sans MS

Aharoni

CASTELLAR

Specify alternate font faces

A visitor will only be able to see your font if they have that font installed on their computer. So, it is possible to specify two or more font face alternatives by listing the font face names, separated by a comma.

```
<font face="Arial, Helvetica">
<font face="Lucida Calligraphy, Comic Sans MS, Lucida Console">
```

When your page is loaded, their browser will display the first font face available. If none of the given fonts are installed, then it will display the default font face Times New Roman.

Setting Font Color

You can set any font color you like using color attribute. You can specify the color that you want by either the color name or hexadecimal code for that color.

```
<html>
<head>
<title>Setting Font Color</title>
```

```
</head>
<body>
<font color="#FF00FF">This text is in pink</font><br />
<font color="red">This text is red</font>
</body>
</html>
```

The <basefont> Element:

The <basefont> element is supposed to set a default font size, color, and typeface for any parts of the document that are not otherwise contained within a <font> tag. You can use the <font> elements to override the <basefont> settings.

The <basefont> tag also takes color, size and face attributes and it will support relative font setting by giving size a value of +1 for a size larger or -2 for two sizes smaller.

```
<html>
<head>
<title>Setting Basefont Color</title>
</head>
<body>
<basefont face="arial, verdana, sans-serif" size="2" color="#ff0000">
<p>This is the page's default font.</p>
<h2>Example of the basefont Element</h2>
<p><font size="+2" color="darkgray">
This is darkgray text with two sizes larger
```

```
</font></p>
<p><font face="courier" size="-1" color="#000000">
It is a courier font, a size smaller and black in color.
</font></p>
</body>
</html>
```

CHAPTER 4

HTML Text Arrangement

4.1. LISTS

HTML offers web authors to arrange text as lists. There are three ways for specifying lists of information. All lists must contain one or more list elements. Lists may contain:

1. An unordered list.
2. An ordered list.
3. A definition list.

4.1.1. Unordered Lists

An unordered list is a collection of related items that have no special order or sequence.

This list is created by using HTML **<ul>** tag. Each item in the list is marked with a bullet.

Example

```
<html>
<head>
<title>HTML Unordered List</title>
</head>
<body>
<ul>
<li>Beetroot</li>
<li>Ginger</li>
<li>Potato</li>
```

```
<li>Radish</li>
</ul>
</body>
</html>
```

This will produce the following result:

- Beetroot
- Ginger
- Potato
- Radish

You can use type attribute for <ul> tag to specify the type of bullet you like. By default, it is a disc. Following are the possible options:

```
<ul type="square">
<ul type="disc">
<ul type="circle">
```

4.1.2. Ordered Lists

If you are required to put your items in a numbered list instead of bulleted, then HTML ordered list will be used. This list is created by using <ol> tag. The numbering starts at one and is incremented by one for each successive ordered list element tagged with <li>.

Example

```
<html>
<head>
<title>HTML Unordered List</title>
</head>
<body>
```

```
<ol>
<li>Beetroot</li>
<li>Ginger</li>
<li>Potato</li>
<li>Radish</li>
</ol>
</body>
</html>
```

This will produce the following result:

1. Beetroot
2. Ginger
3. Potato
4. Radish

You can use **type** attribute for <ol> tag to specify the type of numbering you like. By default, it is a number. Following are the possible options:

```
<ol type="1"> - Default-Case Numerals.
<ol type="I"> - Upper-Case Numerals.
<ol type="i"> - Lower-Case Numerals.
<ol type="a"> - Lower-Case Letters.
<ol type="A"> - Upper-Case Letters.
```

You can use start attribute for <ol> tag to specify the starting point of numbering you need. Following are the possible options:

```
<ol type="1" start="4">    - Numerals starts with 4.
<ol type="I" start="4">    - Numerals starts with IV.
<ol type="i" start="4">    - Numerals starts with iv.
```

```
<ol type="a" start="4">    - Letters starts with d.
<ol type="A" start="4">    - Letters starts with D.
```

4.1.3. Definition Lists

HTML and XHTML supports a list style which is called definition lists where entries are listed like in a dictionary or encyclopedia. The definition list is the ideal way to present a glossary, list of terms, or other name/value list. Definition List makes use of following three tags.

- <dl> - Defines the start of the list
- <dt> - A term
- <dd> - Term definition
- </dl> - Defines the end of the list

Example

```
<html>
<head>
<title>HTML Definition List</title>
</head>
<body>
<dl>
<dt><b>HTML</b></dt>
<dd>This stands for Hyper Text Markup Language</dd>
<dt><b>HTTP</b></dt>
<dd>This stands for Hyper Text Transfer Protocol</dd>
</dl>
</body>
```

```
</html>
```

This will produce the following result:

```
HTML

    This stands for Hyper Text Markup Language

HTTP

    This stands for Hyper Text Transfer Protocol
```

4.2. TABLES

The HTML tables allow web authors to arrange data like text, images, links, other tables, etc. into rows and columns of cells. The HTML tables are created using the <table> tag in which the <tr> tag is used to create table rows and <td> tag is used to create data cells.

Example

```
<html>
<head>
<title>HTML Tables</title>
</head>
<body>
<table border="1">
<tr>
<td>Row 1, Column 1</td>
<td>Row 1, Column 2</td>
</tr>
<tr>
<td>Row 2, Column 1</td>
<td>Row 2, Column 2</td>
</tr>
</table>
</body>
</html>
```

This will produce the following result:

Row 1, Column 1	Row 1, Column 2
Row 2, Column 1	Row 2, Column 2

Here, the border is an attribute of <table> tag and it is used to put a border across all the cells. If you do not need a border, then you can use border="0".

4.2.1. Table Heading

Table heading can be defined using <th> tag. This tag will be put to replace <td> tag, which is used to represent actual data cell. Normally you will put your top row as table heading as shown below, otherwise you can use <th> element in any row.

Example

```
<html>
<head>
<title>HTML Table Header</title>
</head>
<body>
<table border="1">
<tr>
<th>Name</th>
<th>Salary</th>
</tr>
<tr>
```

```
<td>Ramesh Raman</td>
<td>5000</td>
</tr>
<tr>
<td>Shabbir Hussein</td>
<td>7000</td>
</tr>
</table>
</body>
</html>
```

This will produce the following result:

Name	Salary
Ramesh Raman	5000
Shabbir Hussein	7000

4.2.2. Cellpadding and Cellspacing Attributes

There are two attributes called **cellpadding** and **cellspacing** which you will use to adjust the white space in your table cells. The **cellspacing** attribute defines the width of the border, while **cellpadding** represents the distance between cell borders and the content within a cell.

```
<table border="1" cellpadding="5" cellspacing="5">
```

This will produce the following result:

Name	Salary
Ramesh Raman	5000
Shabbir Hussein	7000

4.2.3. Colspan and Rowspan Attributes

You will use **colspan** attribute if you want to merge two or more columns into a single column. Similar way you will use **rowspan** if you want to merge two or more rows.

Example

```
<html>
<head>
<title>HTML Table Colspan/Rowspan</title>
</head>
<body>
<table border="1">
<tr>
<th>Column 1</th>
<th>Column 2</th>
<th>Column 3</th>
</tr>
<tr><td rowspan="2">Row 1 Cell 1</td><td>Row 1 Cell 2</td><td>Row 1 Cell 3</td></tr>
<tr><td>Row 2 Cell 2</td><td>Row 2 Cell 3</td></tr>
```

```
<tr><td colspan="3">Row 3 Cell 1</td></tr>
</table>
</body>
</html>
```

This will produce the following result:

Column 1	Column 2	Column 3
Row 1 Cell 1	Row 1 Cell 2	Row 1 Cell 3
	Row 2 Cell 2	Row 2 Cell 3
Row 3 Cell 1		

4.2.4. Tables Backgrounds

You can set table background using one of the following two ways:

- **bgcolor** attribute - You can set background color for whole table or just for one cell.
- **background** attribute - You can set background image for whole table or just for one cell.

You can also set border color also using **bordercolor** attribute

```
<table border="1" bordercolor="green" bgcolor="yellow">
```

Here is an example of using background attribute. Here we will use an image available in /images directory.

```
<table border="1" bordercolor="green" background= "test.png">
```

4.2.5. Table Height and Width

You can set a table width and height using width and height attributes. You can specify table width or height in terms of pixels or in terms of percentage of available screen area.

```
<table border="1" width="400" height="150">
```

This will produce the following result:

Row 1, Column 1	Row 1, Column 2
Row 2, Column 1	Row 2, Column 2

4.2.6. Table Caption

The caption tag will serve as a title or explanation for the table and it shows up at the top of the table. This tag is deprecated in newer version of HTML/XHTML.

```
<table border="1" width="100%">

<caption>This is the caption</caption>
```

This will produce the following result:

This is the caption	
row 1, column 1	row 1, column 2
row 2, column 1	row 2, column 2

4.2.7. Table Header, Body, and Footer

Tables can be divided into three portions: a header, a body, and a foot. The head and foot are rather similar to headers and footers in a word-processed document that remain the same for every page, while the body is the main content holder of the table.

The three elements for separating the head, body, and foot of a table are:

- <thead> - to create a separate table header.
- <tbody> - to indicate the main body of the table.
- <tfoot> - to create a separate table footer.

A table may contain several <tbody> elements to indicate different pages or groups of data. But it is notable that <thead> and <tfoot> tags should appear before <tbody>

Example:

```
<html>
<head>
<title>HTML Table</title>
</head>
<body>
<table border="1" width="100%">
<thead>
<tr>
<td colspan="4">This is the head of the table</td>
</tr>
</thead>
<tfoot>
<tr>
<td colspan="4">This is the foot of the table</td>
</tr>
```

```
</tfoot>
<tbody>
<tr>
<td>Cell 1</td>
<td>Cell 2</td>
<td>Cell 3</td>
<td>Cell 4</td>
</tr>
</tbody>
</table>
</body>
</html>
```

CHAPTER 5

HTML IMAGES & LINKS

5.1. IMAGES

Images are very important to beautify as well as to depict many complex concepts in simple way on your web page. We will take you through simple steps to use images in your web pages.

5.1.1. Insert Image

You can insert any image in your web page by using <img> tag. Following is the simple syntax to use this tag.

```
<img src="Image URL" ... attributes-list/>
```

The <img> tag is an empty tag, which means that, it can contain only list of attributes and it has no closing tag.

Example

To try following example, let's keep our HTML file test.htm and image file test.png in the same directory:

```
<html>
<head>
<title>Using Image in Webpage</title>
</head>
<body>
<p>Simple Image Insert</p>
<img src="test.png" alt="Test Image" />
</body>
</html>
```

You can use PNG, JPEG or GIF image file based on your comfort but make sure you specify correct image file name in src attribute. Image name is always case sensitive.

The alt attribute is a mandatory attribute which specifies an alternate text for an image, if the image cannot be displayed.

5.1.2. Set Image Location

Usually we keep all the images in a separate directory. So let's keep HTML file test.htm in our home directory and create a subdirectory images inside the home directory where we will keep our image test.png.

```
<img src="images/test.png" alt="Test Image" />
```

Set Image Width/Height

You can set image width and height based on your requirement using width and height attributes. You can specify width and height of the image in terms of either pixels or percentage of its actual size.

```
<img src="test.png" alt="Test Image" width="150" height="100"/>
```

Set Image Border

By default, image will have a border around it, you can specify border thickness in terms of pixels using border attribute. A thickness of 0 means, no border around the picture.

```
<img src="test.png" alt="Test Image" border="3"/>
```

Set Image Alignment

By default, image will align at the left side of the page, but you can use align attribute to set it in the center or right.

```
<img src="test.png" alt="Test Image" border="3" align="right"/>
```

5.2. EMBED MULTIMEDIA

Sometimes you need to add music or video into your web page. The easiest way to add video or sound to your web site is to include the special HTML tag called <embed>. This tag causes the browser itself to include controls for the multimedia automatically provided browser supports <embed> tag and given media type.

You can also include a <noembed> tag for the browsers which don't recognize the <embed> tag. You could, for example, use <embed> to display a movie of your choice, and <noembed> to display a single JPG image if browser does not support <embed> tag.

Example

Here is a simple example to play an embedded midi file:

```
<html>
<head>
<title>HTML embed Tag</title>
</head>
<body>
<embed src="/html/yourfile.mid" width="100%" height="60" >
```

```
    <noembed><img src="yourimage.gif" alt="Alternative Media" ></noembed>
   </embed>
   </body>
   </html>
```

5.2.1.The <embed> Tag Attributes

Following is the list of important attributes which can be used with <embed> tag.

- **align** Determines how to align the object. It can be set to either center, left or right.
- **autostart** This boolean attribute indicates if the media should start automatically. You can set it either true or false.
- **loop** Specifies if the sound should be played continuously (set loop to true), a certain number of times (a positive value) or not at all (false)
- **playcount** Specifies the number of times to play the sound.
- **hidden** Specifies if the multimedia object should be shown on the page. A false value means no and true values means yes.
- **width** Width of the object in pixels
- **height** Height of the object in pixels
- **name** A name used to reference the object.
- **src** URL of the object to be embedded.
- **volume** Controls volume of the sound. Can be from 0 (off) to 100 (full volume).

5.3. LINKS

A webpage can contain various links that take you directly to other pages and even specific parts of a given page. These links are known as hyperlinks.

Hyperlinks allow visitors to navigate between Web sites by clicking on words, phrases, and images. Thus you can create hyperlinks using text or images available on a webpage.

5.3.1.Linking Documents

A link is specified using HTML tag <a>. This tag is called anchor tag and anything between the opening <a> tag and the closing tag becomes part of the link and a user can click that part to reach to the linked document. Following is the simple syntax to use <a> tag.

```
<a href="Document URL" ... attributes-list>Link Text</a>
```

Example

Let's try following example which links http://www.yahoo.com at your page:

```
<html>
<head>
<title>Hyperlink Example</title>
</head>
<body>
<p>Click following link</p>
<a href="http://www.yahoo.com" target="_self"> Yahoo </a>
</body>
```

```
</html>
```

This will produce the following result, where you can click on the link generated to reach to the home page of Yahoo (in this example).

Click following link

```
Yahoo
```

5.3.2.The target Attribute

We have used target attribute in our previous example. This attribute is used to specify the location where linked document is opened. Following are the possible options:

- _blank : Opens the linked document in a new window or tab.
- _self : Opens the linked document in the same frame.
- _parent : Opens the linked document in the parent frame.
- _top : Opens the linked document in the full body of the window.

Example

```
<a href="http://www.yahoo.com" target="_blank">Opens in New</a> |
<a href="http://www.yahoo.com" target="_self">Opens in Self</a> |
<a href="http://www.yahoo.com" target="_parent">Opens in Parent</a> |
<a href="http://www.yahoo.com" target="_top">Opens in Body</a>
```

This will produce the following result, where you can click on different links to understand the difference between various options given for target attribute.

Click any of the following links

```
Opens in New | Opens in Self | Opens in Parent | Opens in Body
```

5.3.3.Use of Base Path

When you link HTML documents related to the same website, it is not required to give a complete URL for every link. You can get rid of it if you use <base> tag in your HTML document header. This tag is used to give a base path for all the links. So your browser will concatenate given relative path to this base path and will make a complete URL.

Example

```
<html>
<head>
<title>Hyperlink Example</title>
<base href="http://www.yahoo.com">
</head>
```

5.3.4.Internal Links

Links can also be created inside large documents to simplify navigation. Today's world wants to be able to get the information quickly. Internal links can help you meet these goals.

Select some text at a place in the document that you would like to create a link to, then add an anchor to link to like this:

```
<A NAME="bookmark_name"></A>
```

The Name attribute of an anchor element specifies a location in the document that we link to shortly. All NAME attributes in a document must be unique.

Next select the text that you would like to create as a link to the location created above.

```
<A HREF="#bookmark_name">Go To  Book Mark</A>
```

5.3.5.IMAGE LINKS

We have seen how to create hypertext link using text and we also learnt how to use images in our webpages. Now, we will learn how to use images to create hyperlinks.

Example

It's simple to use an image as hyperlink. We just need to use an image inside hyperlink at the place of text as shown below:

```
<html>
<head>
<title>Image Hyperlink Example</title>
</head>
<body>
<p>Click following link</p>
<a href=" http://www.yahoo.com " target="_self">
  <img src="logo.png" alt="Yahoo" border="0"/>
</a>
</body>
</html>
```

5.3.6.E-mail Link

It is not difficult to put an HTML email link on your webpage but it can cause unnecessary spamming problem for your email account. There

are people, who can run programs to harvest these types of emails and later use them for spamming in various ways.

You can have another option to facilitate people to send you emails. One option could be to use HTML forms to collect user data and then use PHP or CGI script to send an email.

A simple example, check our Contact Us Form. We take user feedback using this form and then we are using one CGI program which is collecting this information and sending us email to the one given email ID.

5.3.7.HTML Email Tag

HTML <a> tag provides you option to specify an email address to send an email. While using <a> tag as an email tag, you will use mailto: email address along with href attribute. Following is the syntax of using mailto instead of using http.

```
<a href= "mailto: abc@example.com">Send Email</a>
```

This code will generate the following link which you can use to send email.

```
Send Email
```

Now, if a user clicks this link, it launches one Email Client (like Lotus Notes, Outlook Express etc.) installed on your user's computer. There is another risk to use this option to send email because if user do not have email client installed on their computer then it would not be possible to send email.

CHAPTER 6

HTML FRAMES

HTML frames are used to divide your browser window into multiple sections where each section can load a separate HTML document. A collection of frames in the browser window is known as a frameset. The window is divided into frames in a similar way the tables are organized: into rows and columns.

6.1. Creating Frames

To use frames on a page we use <frameset> tag instead of <body> tag. The <frameset> tag defines, how to divide the window into frames. The rows attribute of <frameset> tag defines horizontal frames and cols attribute defines vertical frames. Each frame is indicated by <frame> tag and it defines which HTML document shall open into the frame.

Example

Following is the example to create three horizontal frames:

```
<html>
<head>
<title>HTML Frames</title>
</head>
<frameset rows="10%,80%,10%">
  <frame name="top" src="/html/top_frame.htm" />
  <frame name="main" src="/html/main_frame.htm" />
  <frame name="bottom" src="/html/bottom_frame.htm" />
  <noframes>
  <body>
```

```
Your browser does not support frames.
  </body>
  </noframes>
</frameset>
</html>
```

This will produce the following result:



Example

Let's put the above example as follows, here we replaced rows attribute by cols and changed their width. This will create all the three frames vertically:

```
<frameset cols="25%,50%,25%">
```

This will produce the following result:



6.1.1. The <frameset> Tag Attributes

Following are important attributes of the <frameset> tag:

- **Cols**: Specifies how many columns are contained in the frameset and the size of each column. You can specify the width of each column in one of the four ways:
  - Absolute values in pixels. For example, to create three vertical frames, use cols="100, 500,100".
  - A percentage of the browser window. For example, to create three vertical frames, use cols="10%, 80%,10%". Using a wildcard symbol. For example, to create three vertical frames, use cols="10%, *,10%". In this case wildcard takes remainder of the window.
  - As relative widths of the browser window. For example, to create three vertical frames, use cols="3*,2*,1*". This is an alternative to percentages. You can use relative widths of the browser window. Here the window is divided into sixths: the first column takes up half of the window, the second takes one third, and the third takes one sixth.
- **Rows**: This attribute works just like the cols attribute and takes the same values, but it is used to specify the rows in the frameset. For example, to create two horizontal frames, use rows="10%, 90%". You can specify the height of each row in the same way as explained above for columns.
- **border**: This attribute specifies the width of the border of each frame in pixels. For example, border="5". A value of zero means no border.
- **frameborder**: This attribute specifies whether a three-dimensional border should be displayed between frames. This attribute takes value either 1 (yes) or 0 (no). For example frameborder="0" specifies no border.

- **framespacing:** This attribute specifies the amount of space between frames in a frameset. This can take any integer value. For example framespacing="10" means there should be 10 pixels spacing between each frames.

6.1.2. The <frame> Tag Attributes

Following are the important attributes of <frame> tag:

- **src**: This attribute is used to give the file name that should be loaded in the frame. Its value can be any URL. For example, src="/html/top_frame.htm" will load an HTML file available in html directory.
- **name**: This attribute allows you to give a name to a frame. It is used to indicate which frame a document should be loaded into. This is especially important when you want to create links in one frame that load pages into an another frame, in which case the second frame needs a name to identify itself as the target of the link.
- **frameborder:** This attribute specifies whether or not the borders of that frame are shown; it overrides the value given in the frameborder attribute on the <frameset> tag if one is given, and this can take values either 1 (yes) or 0 (no).
- **marginwidth:** This attribute allows you to specify the width of the space between the left and right of the frame's borders and the frame's content. The value is given in pixels. For example marginwidth="10".
- **marginheight:** This attribute allows you to specify the height of the space between the top and bottom of the frame's borders and its contents. The value is given in pixels. For example marginheight="10".

- **noresize:** By default, you can resize any frame by clicking and dragging on the borders of a frame. The noresize attribute prevents a user from being able to resize the frame. For example noresize="noresize".
- **scrolling:** This attribute controls the appearance of the scrollbars that appear on the frame. This takes values either "yes", "no" or "auto". For example scrolling="no" means it should not have scroll bars.
- **longdesc:** This attribute allows you to provide a link to another page containing a long description of the contents of the frame. For example longdesc="framedescription.htm"

6.1.3. Frame's name and target attributes

One of the most popular uses of frames is to place navigation bars in one frame and then load main pages into a separate frame.

Let's see following example where a test.htm file has following code:

```
<html>
<head>
<title>HTML Target Frames</title>
</head>
<frameset cols="200, *">
  <frame src="/html/menu.htm" name="menu_page" />
  <frame src="/html/main.htm" name="main_page" />
  <noframes>
  <body>
    Your browser does not support frames.
  </body>
```

```
    </noframes>
  </frameset>
  </html>
```

Here, we have created two columns to fill with two frames. The first frame is 200 pixels wide and will contain the navigation menu bar implemented by menu.htm file. The second column fills in remaining space and will contain the main part of the page and it is implemented by main.htm file. For all the three links available in menu bar, we have mentioned target frame as main_page, so whenever you click any of the links in menu bar, available link will open in main page.

Following is the content of menu.htm file

```
<html>
<body bgcolor="#4a7d49">
<a href="http://www.google.com" target="main_page">Google</a>
<br /><br />
<a href="http://www.microsoft.com" target="main_page">Microsoft</a>
<br /><br />
<a href="http://news.bbc.co.uk" target="main_page">BBC News</a>
</body>
</html>
```

Following is the content of main.htm file:

```
<html>
<body bgcolor="#b5dcb3">
```

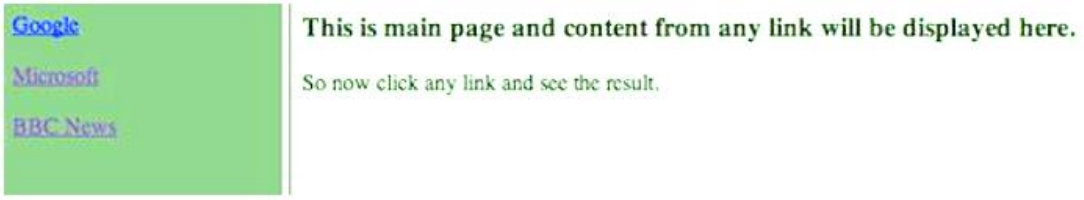
```
<h3>This is main page and content from any link will be displayed here.</h3>

<p>So now click any link and see the result.</p>

</body>

</html>
```

When we load test.htm file, it produces following result:



CHAPTER 7

HTML FORMS

7.1. Forms

HTML Forms are required, when you want to collect some data from the site visitor. For example, during user registration you would like to collect information such as name, email address, credit card, etc.

A form will take input from the site visitor and then will post it to a back-end application such as CGI, ASP Script or PHP script etc. The back-end application will perform required processing on the passed data based on defined business logic inside the application.

There are various form elements available like text fields, textarea fields, drop-down menus, radio buttons, checkboxes, etc.

The HTML <form> tag is used to create an HTML form and it has following syntax:

```
<form action="Script URL" method="GET|POST">

   form elements like input, textarea etc.

</form>
```

7.1.1. Form Attributes

Apart from common attributes, following is a list of the most frequently used form attributes:

- **action** Backend script ready to process your passed data.
- **method** Method to be used to upload data. The most frequently used are GET and POST methods.
- **target** Specify the target window or frame where the result of the script will be displayed. It takes values like _blank, _self, _parent etc.

- **enctype** You can use the enctype attribute to specify how the browser encodes the data before it sends it to the server. Possible values are:
    - application/x-www-form-urlencoded - This is the standard method most forms use in simple scenarios.
    - mutlipart/form-data - This is used when you want to upload binary data in the form of files like image, word file etc.

7.1.2. HTML Form Controls

There are different types of form controls that you can use to collect data using HTML form:

- Text Input Controls
- Checkboxes Controls
- Radio Box Controls
- Select Box Controls
- File Select boxes
- Hidden Controls
- Clickable Buttons
- Submit and Reset Button

Text Input Controls

There are three types of text input used on forms:

- Single-line text input controls - This control is used for items that require only one line of user input, such as search boxes or names. They are created using HTML <input> tag.

- Password input controls - This is also a single-line text input but it masks the character as soon as a user enters it. They are also created using HTMl <input> tag.
- Multi-line text input controls - This is used when the user is required to give details that may be longer than a single sentence. Multi-line input controls are created using HTML <textarea> tag.

Single-line text input controls

This control is used for items that require only one line of user input, such as search boxes or names. They are created using HTML <input> tag.

Example

Here is a basic example of a single-line text input used to take first name and last name:

```
<html>
<head>
<title>Text Input Control</title>
</head>
<body>
<form >
First name:  <input type="text" name="first_name" />
<br> Last name:  <input type="text" name="last_name" />
</form>
</body>
```

```
</html>
```

Attributes

Following is the list of attributes for <input> tag for creating text field.

- **type** Indicates the type of input control and for text input control it will be set totext.
- **name** Used to give a name to the control which is sent to the server to be recognized and get the value.
- **value** This can be used to provide an initial value inside the control.
- **size** Allows to specify the width of the text-input control in terms of characters.
- **maxlength** Allows to specify the maximum number of characters a user can enter into the text box.

Password Input controls

This is also a single-line text input but it masks the character as soon as a user enters it. They are also created using HTML <input> tag but type attribute is set to password.

Example

Here is a basic example of a single-line password input used to take user password:

```
<form >
User ID :  <input type="text" name="user_id" /><br>
Password:  <input type="password" name="password" />
</form>
```

Multiple-Line Text Input Controls

This is used when the user is required to give details that may be longer than a single sentence. Multi-line input controls are created using HTML <textarea> tag.

```
<form>
Description: <br />
<textarea rows="5" cols="50" name="description">
Enter description here...
</textarea>
</form>
```

This will produce the following result:



Attributes

Following is the list of attributes for <textarea> tag.

- **name** Used to give a name to the control which is sent to the server to be recognized and get the value.
- **rows** Indicates the number of rows of text area box.
- **cols** Indicates the number of columns of text area box

Checkbox Control

Checkboxes are used when more than one option is required to be selected. They are also created using HTML <input> tag but type attribute is set to checkbox.

```
<form>
<input type="checkbox" name="maths" value="on"> Maths
<input type="checkbox" name="physics" value="on"> Physics
</form>
```

Attributes

Following is the list of attributes for <checkbox> tag.

- **type** Indicates the type of input control and for checkbox input control it will be set to checkbox.
- **name** Used to give a name to the control which is sent to the server to be recognized and get the value.
- **value** The value that will be used if the checkbox is selected.
- **checked** Set to checked if you want to select it by default.

Radio Button Control

Radio buttons are used when out of many options, just one option is required to be selected. They are also created using HTML <input> tag but type attribute is set to radio.

```
<form>
<input type="radio" name="subject" value="maths"> Maths
<input type="radio" name="subject" value="physics"> Physics
</form>
```

Select Box Control

A select box, also called drop down box which provides option to list down various options in the form of drop down list, from where a user can select one or more options.

```
<form>
<select name="dropdown">
<option value="Maths" selected>Maths</option>
<option value="Physics">Physics</option>
</select>
</form>
```

Attributes

- **name** Used to give a name to the control which is sent to the server to be recognized and get the value.
- **size** This can be used to present a scrolling list box.
- **multiple** If set to "multiple" then allows a user to select multiple items from the menu.
- **value** The value that will be used if an option in the select box is selected.
- **selected** Specifies that this option should be the initially selected value when the page loads.
- **label** An alternative way of labeling options

Button Controls

There are various ways in HTML to create clickable buttons. You can also create a clickable button using <input> tag by setting its type attribute to button. The type attribute can take the following values:

- **submit** This creates a button that automatically submits a form.
- **reset** This creates a button that automatically resets form controls to their initial values.
- **button** This creates a button that is used to trigger a client-side script when the user clicks that button.
- **image** This creates a clickable button but we can use an image as background of the button.

Example

```
<html>
<head>
<title>File Upload Box</title>
</head>
<body>
<form>
<input type="submit" name="submit" value="Submit" />
<input type="reset" name="reset"  value="Reset" />
<input type="button" name="ok" value="OK"  />
<input type="image" name="imagebutton" src="/html/images/logo.png" />
</form>
</body>
</html>
```